33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F13-SR-002
415V BALL MILL SWITCHROOM

F13-TR-003
TRANSFORMER ROOM

F13-TR-003
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
4000/ 1A
2.5VA
5P20

F13-SW-003
BALL MILL 415V MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
5VA
5P20
00F-01
4000/ 5A
30VA
C10.5
00F-08
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
AFD
ACB
2A
4000A
50G
50
51
51G
24VDC CONTROL POWER SUPPLY
NOTE 7
125A
160A
SD
NOTE 6

NORTHERN STAR RESOURCES LIMITED

| | |
|---|---|
| Approved | X |
| Approved as Noted | |
| Not Approved | |
| For Information Only | |

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

415V 3NE~ 50Hz 4000A 65kA for 1sec
INC

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6F-04 | 6F-01 | 5R-01 | 3F-01 | 1F-09 | 1F-13 | 7F-01 | 1F-17 | 1F-21 | 2F-01 | 2F-05 | 2F-09 | |
| 63A | 100A | 20A | 20A | 32A | 32A | 800A | 32A | 32A | 32A | 32A | 32A | |
| 70A | 90A | 70A | 70A | 70A | 70A | 400A | 70A | 70A | 70A | 70A | 70A | |
| | | | EOL | EOL | EOL | | EOL | EOL | EOL | EOL | EOL | |
| CP | CP | CP | M 3~ | M 3~ | M 3~ | | M 3~ | M 3~ | M 3~ | M 3~ | M 3~ | |
| 20kW | 29kW | 6kW | 0.37kW | 7.5kW | 7.5kW | | 7.5kW | 7.5kW | 7.5kW | 7.5kW | 7.5kW | |
| F13-BLC-402 BALL MILL BALL CHARGER 2000-F13-SCM-EL-10909 | F13CU408-CP BALL MILL VSD COOLING SYSTEM CONTROL PANEL 2000-F13-SCM-EL-10912 | F13CU409-CP CYCLONE FEED PUMP WATER COOLER UNIT CONTROL PANEL 2000-F13-SCM-EL-10942 | F13-FN-426 BALL MILL MOTOR LUBE SKID COOLER FAN 2000-F13-SCM-EL-10913 | F13-FN-435 BALL MILL MOTOR 1 FAN 1 2000-F13-SCM-EL-10914 | F13-FN-436 BALL MILL MOTOR 1 FAN 2 2000-F13-SCM-EL-10915 | | F13-FN-437 BALL MILL MOTOR 1 FAN 3 2000-F13-SCM-EL-10916 | F13-FN-438 BALL MILL MOTOR 1 FAN 4 2000-F13-SCM-EL-10917 | F13-FN-439 BALL MILL MOTOR 2 FAN 1 2000-F13-SCM-EL-10918 | F13-FN-440 BALL MILL MOTOR 2 FAN 2 2000-F13-SCM-EL-10919 | F13-FN-441 BALL MILL MOTOR 2 FAN 3 2000-F13-SCM-EL-10920 | |

FEEDER TO F13-SW-103
2000-F13-SCM-EL-10952

E-SHUT
F13SR002-LCS
NOTE 12

CONTINUED ON SHT 2
2000-F13-SLD-EL-10902

## NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F13-SW-003 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F13-BLK-IC-10002.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

## LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA ST TRIP SET POINT / xxxA CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

# 2000-F13-SLD-EL-10901_Rev1

Ball mill F13-SW-003 415V MCC single line diagram sheet 1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\235117 KCGM GROWTH PROJECT\03 DELIVERABLES\ELV\F13 - MILLING & CLASSIFICATION\2000-F13-SLD-EL-10901.DWG 25SEP25 12:52PM HOTSEA.T04